My research interest lies in **Natural Language Processing (NLP) with a focus on explainability and efficiency**. More specifically, I am interested in exploring the faithfulness of explanations and creating more efficient and accessible NLP. Below, I describe some questions I raised from my projects done at the University of Washington with Ana Marasović and Noah A. Smith.

1. Given an interpretation method, to what degree do generated explanations show the reasoning process of the corresponding models? (i.e. How faithful are these explanations?)

2. With a sufficiently faithful interpretation method, how can we develop a computationally efficient NLP system?

Evaluating the level of faithfulness of interpretation methods While working as a data scientist in industry, I realized how crucial an interpretation method will be to a practitioner to understand model's internals and make design options. While these methods can aid model deployment and understanding for practitioners, sometimes the generated explanations do not accurately reflect the decision-making process of the underlying model [10] and lead to risks of misinforming end-users [5]. In this case, having a reliable measure of the level of faithfulness can help closer the distance between plausibility (how convincing the explanations are to humans) and faithfulness (how much the explanations show the reasoning process of models), thus guiding practitioners to make better decisions and reduce issues caused by unreliable predictions especially in some critical applications such as health care and policymaking.

As a starting point, I conducted my first NLP research project [11] published in Findings of ACL 2021 with Ana Marasović. In this work, I analyzed and compared the interpretations presented by effective attention [3] and self-attention [12]. Effective attention appears in a mild condition, in which self-attention can be decomposed into two components. One of the components does not affect the prediction result since its product with the value matrix is zero. The other part, "effective attention", is the part that contributes to the model output. We find that effective attention is less associated with the features related to the language modeling pretraining such as the separator token and has more potential to illustrate linguistic features captured by the model for solving the end task [11]. The analysis method we used, however, solely looks at the attention weights, which are proved to be not the only factor that determines the output of self-attention [7], thus leading me to question the level at which effective attention uncovers the model's decision process and how does it compare to the other factor mentioned in [7].

Motivated by the findings of my project, I started to think about the question, **when proposing an interpretation method, how do we assess the level of faithfulness of the produced explanations?** After a series of reading, I learned that there are several prior works on measuring the degree of faithfulness either by observing the result when perturbing the tokens that are identified as important by the method [4, 6, 8], or by setting some desired properties for a hypothesized sufficiently faithful explanations [2, 9, 14]. While exploring, another question goes into my mind: if an evaluation method is not assessing the faithfulness thoroughly, we may encounter the same issue when evaluating interpretation methods - how much does this evaluation method tell us about the level of faithfulness interpretation methods? The first idea that comes into my mind is to avoid evaluating explanations by developing models that are interpretable by design. But **when can we claim that some models are "interpretable"**, especially given that we have seen the example of self-attention? Another direction that comes into my mind targets interpretation methods: can we develop interpretation methods that have strong theoretical support and can expressively approximate the underlying model's decision-making process? This leads us back to the question of evaluating interpretation methods. However, in many scenarios, we can empirically examine how much the interpretation methods tell us about the system within a specific domain and model, but when can we escape from this scheme and extend the scope of this evaluation method to a broader context?

Research towards these questions can not only benefit practitioners in making decisions but also give us a better understanding of the models and shed light on making better NLP with these methods. Therefore, my interests start from but will not end at these questions. Eventually, I would like to apply what I have developed to building more accessible and faithfully interpretable NLP.

Ameliorating computational efficiency with interpretable approaches In some cases, the interpretation methods can be too slow to produce explanations. This impedes researchers without a decent amount of computational resources from working on more in-depth analyses with these methods.

While working on analyzing effective attention, I noticed that the Singular Value Decomposition (SVD) employed for computing effective attention slows down the training process compared to the case of standard attention.

Therefore, we attempted to combine effective attention with Linformer [13], a linear-time transformer built on the idea of low-rank approximation of attention matrix, in order to speed up the process of computing effective attention. However, I soon realized that approximating attention matrix with a low-rank matrix before computing effective attention could damage the advantage of interpretability brought by effective attention, as the assumption that standard attention matrix being the pairwise importance scores between tokens failed to hold due to a linear transformation in Linformer.

Similarly, I experienced the same **trade-off between efficiency and interpretability** on my second project, which aims at improving the asymptotic complexity of computing the attention flow [1]. Attention flow is an interpretation method that intends to quantify the flow of information within layers of Transformer by computing the $s-t$ flows between tokens in different layers, in which tokens at the first layer are considered as sources (s) while other tokens are considered as sinks (t). Because a large amount of $s-t$ flow computations are required to compute attention flow, and there is no deterministic algorithm that reaches a desired asymptotic performance for $s-t$ flow, I intended to speedup the attention flow computation with two methods: pruning out the vertices with presumably negligible flow values in earlier layers to increase the speed and designing a randomized algorithm particularly for attention flow to decrease the asymptotic time complexity. Nevertheless, the first method failed to give an approximation that is distinguishable with a random matrix generated from a normal distribution of attention flow; the second method failed to preserve the original hypothesis of attention flow being the flow of information within Transformer. When trying to understand why this is the case, I realized that both of the proposed methods require strong hypotheses and may damage the faithfulness of explanations produced by the original method.

Seeing how efficiency can obstruct researchers without many computational resources from working on these interpretation methods and how efficiency may conflict with interpretability, **I am stimulated to work on enhancing computational efficiency with less or without a decrease in the interpretability of interpretation methods.** Furthermore, I would like to also work towards improving training efficiency with a sufficiently faithful interpretation, since low efficiency during training impairs more accessibility to NLP research and conveys more reproducibility than low efficiency during validation does.

Motivation to obtain a PhD `[Approximately 2 paragraphs, along with motivation to apply for the specific school and interest with specific professors.]`

References

[1] Samira Abnar and Willem Zuidema. Quantifying attention flow in transformers. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 4190–4197, 2020.

[2] Marco Ancona, Enea Ceolini, Cengiz Öztireli, and Markus Gross. Towards better understanding of gradient-based attribution methods for deep neural networks. *arXiv preprint arXiv:1711.06104*, 2017.

[3] Gino Brunner, Yang Liu, Damian Pascual, Oliver Richter, Massimiliano Ciaramita, and Roger Wattenhofer. On identifiability in transformers. In *International Conference on Learning Representations*, 2019.

[4] Jay DeYoung, Sarthak Jain, Nazneen Fatema Rajani, Eric Lehman, Caiming Xiong, Richard Socher, and Byron C Wallace. Eraser: A benchmark to evaluate rationalized nlp models. *arXiv preprint arXiv:1911.03429*, 2019.

[5] Alon Jacovi and Yoav Goldberg. Towards faithfully interpretable nlp systems: How should we define and evaluate faithfulness? *arXiv preprint arXiv:2004.03685*, 2020.

[6] Sarthak Jain and Byron C Wallace. Attention is not explanation. *arXiv preprint arXiv:1902.10186*, 2019.

[7] Goro Kobayashi, Tatsuki Kuribayashi, Sho Yokoi, and Kentaro Inui. Attention is not only a weight: Analyzing transformers with vector norms. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 7057–7075, 2020.

[8] Sofia Serrano and Noah A Smith. Is attention interpretable? *arXiv preprint arXiv:1906.03731*, 2019.

[9] Avanti Shrikumar, Peyton Greenside, and Anshul Kundaje. Learning important features through propagating activation differences. In *International Conference on Machine Learning*, pp. 3145–3153. PMLR, 2017.

[10] Leon Sixt, Maximilian Granz, and Tim Landgraf. When explanations lie: Why many modified bp attributions fail. In *International Conference on Machine Learning*, pp. 9046–9057. PMLR, 2020.

[11] Kaiser Sun and Ana Marasović. Effective attention sheds light on interpretability. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pp. 4126–4135, Online, August 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.findings-acl.361. URL https://aclanthology.org/2021.findings-acl.361.

[12] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in neural information processing systems*, pp. 5998–6008, 2017.

[13] Sinong Wang, Belinda Z Li, Madian Khabsa, Han Fang, and Hao Ma. Linformer: Self-attention with linear complexity. *arXiv preprint arXiv:2006.04768*, 2020.

[14] Fan Yin, Zhouxing Shi, Cho-Jui Hsieh, and Kai-Wei Chang. On the faithfulness measurements for model interpretations. *arXiv preprint arXiv:2104.08782*, 2021.